

An End-to-End Deep Learning Pipeline for Automated Cancer Histopathology Classification Using the LC25000 Dataset

Amir Aynede ^{1,*}

^{1,*} Department of Pharmacy and Biotechnology - FaBiT, Alma Mater Studiorum - University of Bologna;
amir.aynede@studio.unibo.it

Abstract

Accurate and efficient cancer diagnosis remains a critical challenge in clinical pathology. Manual examination of histopathological slides is time-consuming and prone to inter-observer variability emphasizing the need for automated and interpretable diagnostic systems. This study presents a comprehensive deep learning pipeline for histopathological cancer subtype classification using the publicly available LC25000 dataset which contains 25,000 images across five tissue classes from lung and colon specimens. A transfer learning approach based on a fine-tuned ResNet18 architecture was implemented, supported by automated data preprocessing, augmentation, and reproducible training procedures using the PyTorch framework. Model optimization employed AdamW with learning rate scheduling and early stopping, while performance evaluation used accuracy, precision, recall and F1-score metrics. Interpretability was achieved through Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize class-discriminative regions within tissue samples. The model achieved near-perfect classification performance (accuracy ≈ 1.00) with strong generalization across all histopathological categories. Grad-CAM visualizations confirmed that network attention corresponded to diagnostically relevant cellular structures, supporting clinical interpretability.

Keywords: Histopathology Image Classification; Deep Learning; Explainable Artificial Intelligence; Convolutional Neural Networks (CNNs); Cancer Diagnostics

1. Introduction

Cancer diagnosis through histopathological assessment remains the cornerstone of oncological decision-making. Despite its diagnostic value the manual examination of tissue slides is inherently limited by human subjectivity, time constraints and inter-observer variability (Madabhushi & Lee, 2016; Komura & Ishikawa, 2018). These challenges underscore the need for computational systems capable of delivering reproducible, objective and efficient diagnostic support within digital pathology workflows. Recent advances in deep learning (DL) have demonstrated transformative potential in computational pathology. In particular, convolutional neural networks (CNNs) have achieved state-of-the-art performance in diverse medical imaging tasks by autonomously learning hierarchical tissue representations from raw pixel data (LeCun et al., 2015; Litjens et al., 2017). Such models have shown efficacy in cancer detection, subtype classification, and grading, offering scalability and consistency beyond traditional manual analysis. However, the translation of CNN-based systems into clinical settings

Received: 10/21/2025

Revised: date

Accepted: date

Published: date

Citation: To be added by editorial staff during production.

Copyright: © 2025 by the authors. Submitted for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

remains constrained by insufficient interpretability, lack of methodological standardization, and limited reproducibility. This study addresses these gaps by developing a comprehensive, reproducible and interpretable deep learning pipeline for histopathological image classification using the publicly available LC25000 dataset (Borkowski et al., 2019). The dataset comprises balanced samples from five distinct classes—*lung adenocarcinoma*, *lung squamous cell carcinoma*, *benign lung tissue*, *colon adenocarcinoma*, and *benign colon tissue*—enabling systematic evaluation of deep neural architectures across multiple cancer subtypes. The primary objective of this work is to construct a robust CNN-based framework capable of automated cancer subtype recognition while maintaining interpretability through Gradient-weighted Class Activation Mapping (Grad-CAM)(Selvaraju et al., 2017). The proposed system emphasizes three key dimensions of contribution:

The proposed framework demonstrates reproducibility, interpretability and scalability through its integrated design and clinical relevance. The entire pipeline—from dataset preparation and model training to evaluation and visualization—was developed using a modular and transparent structure, ensuring methodological rigor and reproducible experimentation. Through Grad-CAM-based visualization, the system achieves clinical interpretability by highlighting discriminative regions within histopathological samples allowing verification of the model’s attention on diagnostically meaningful tissue structures. Furthermore, the framework is inherently scalable and adaptable, providing a robust experimental foundation for future extensions such as multi-modal data integration, uncertainty estimation and deployment within clinical decision-support systems. By uniting methodological robustness with interpretability, this work advances the application of explainable deep learning in digital pathology. The proposed pipeline demonstrates that transfer learning-based CNNs can effectively distinguish cancer subtypes in histopathological imagery while providing transparent decision rationales—an essential prerequisite for clinical trust and regulatory compliance in AI-assisted diagnosis.

2. Materials and Methods

2.1. Dataset Description

The experiments were conducted using the publicly available LC25000 dataset (Borkowski et al., 2019), a curated collection of histopathological image patches derived from lung and colon tissue specimens. The dataset comprises 25,000 RGB images, uniformly distributed across five diagnostic categories: *lung adenocarcinoma* (*lung_aca*), *lung squamous cell carcinoma* (*lung_scc*), *benign lung tissue* (*lung_n*), *colon adenocarcinoma* (*colon_aca*), and *benign colon tissue* (*colon_n*). Each image has a spatial resolution of 224×224 pixels, preprocessed to ensure uniformity in illumination and color normalization. The dataset’s balanced structure eliminates class imbalance bias and enables direct comparison of model performance across classes.

2.2. Exploratory Data Analysis (EDA)

An exploratory data analysis (EDA) was conducted as an initial step to evaluate the LC25000 dataset’s composition, visual characteristics and suitability for deep learning. The procedure involved assessing class balance, visual inspection of representative tissue

samples, measurement of brightness distribution, and simulation of augmentation effects. Class frequency plots were generated to verify dataset uniformity while sample grids were used to qualitatively examine morphological variations among *colon* and *lung* tissue subtypes. Image brightness and color intensity distributions were analyzed to detect potential inconsistencies in staining or illumination. In addition, a set of augmentation transformations—including random flips, rotations and color jitter—was applied to examine how artificial variability could enhance model robustness. These EDA steps ensured that the dataset was well-structured, visually coherent and appropriately standardized for convolutional neural network (CNN) training.

2.3. Data Preparation and Partitioning

To ensure experimental rigor and prevent data leakage, the dataset was stratified and divided into three non-overlapping subsets: training (70%), validation (15%), and test (15%). A custom Python script automated the partitioning process by creating separate directory structures (train/, val/, test/) aligned with the folder-based input requirements of `torchvision.datasets.ImageFolder`. Each subset maintained class proportionality to preserve the distribution of all five tissue types. Data loading was implemented through the PyTorch `DataLoader` API with batch normalization and shuffling enabled, ensuring randomized exposure of samples during training.

2.4. Data Augmentation and Preprocessing

To enhance model generalization and mitigate overfitting, data augmentation techniques were applied to the training dataset using `torchvision.transforms`. The augmentation strategy included random horizontal and vertical flips, random rotations within $\pm 15^\circ$, random cropping and resizing to 224×224 pixels and color jitter adjustments to vary brightness and contrast. These transformations increased the diversity of training samples enabling the model to learn more robust and invariant feature representations. Following augmentation, all images were normalized to the ImageNet mean and standard deviation to align with the expected input distribution of pretrained convolutional neural networks. Importantly, no augmentation was applied to the validation and test sets to maintain the integrity of the performance evaluation and ensure unbiased assessment of the model's generalization capability.

2.5. Model Architecture

The classification framework was built upon a pretrained ResNet18 architecture (He et al., 2016), selected for its proven performance in transfer learning tasks and computational efficiency. The pretrained convolutional layers were initialized with ImageNet weights to leverage low-level feature representations such as texture and morphology. The model's final fully connected (FC) layer was replaced with a new dense layer having an output dimension of five units corresponding to the dataset's class count. The modified architecture allowed dynamic adjustment of the output layer to accommodate potential future dataset variations. Batch normalization and ReLU activations were retained throughout the network to stabilize gradient propagation. For comparative analysis, the framework was designed to optionally support other CNN variants such as ResNet34 and EfficientNet-B0 enabling scalability of experimentation.

2.6. Training Configuration

Model training was implemented using the PyTorch deep learning framework. The network was optimized with the AdamW optimizer (Loshchilov & Hutter, 2019) using an initial learning rate of 1×10^{-3} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-4} for regularization. The cross-entropy loss function served as the optimization criterion, appropriate for multi-class classification tasks. A ReduceLROnPlateau scheduler dynamically adjusted the learning rate when validation loss stagnated promoting stable convergence. Training was conducted for a maximum of 30 epochs with an early-stopping mechanism (patience = 5) employed to terminate training once validation performance ceased improving. Preliminary experiments with extended runs (up to 50 epochs) showed that both training and validation losses plateaued around epochs 25–28 indicating convergence. Extending training beyond this point yielded negligible performance gains ($< 0.1\%$) while increasing computational cost and risk of overfitting. Therefore, limiting training to 30 epochs with early stopping provided an optimal balance between efficiency, stability, and generalization. Each epoch recorded real-time training and validation loss, accuracy and F1-score, and the model weights corresponding to the highest validation accuracy were retained for downstream evaluation. All experiments were executed on an NVIDIA T4 GPU environment (Google Colab) ensuring efficient parallel computation and reproducibility.

2.7. Evaluation Metrics

Model performance was evaluated using a comprehensive set of complementary metrics to capture various aspects of classification effectiveness. These included accuracy (ACC), representing the proportion of correctly classified images; precision (P), indicating the positive predictive value per class; recall (R), reflecting class-wise sensitivity; and the F1-score (F_1), calculated as the harmonic mean of precision and recall to balance both measures. Additionally, a confusion matrix was generated to visualize the distribution of true versus predicted class labels and identify potential patterns of misclassification. All metrics were computed using the torchmetrics library for GPU-accelerated, batch-wise evaluation, ensuring high computational efficiency and reproducibility while minimizing manual aggregation errors. For validation and cross-verification, statistical reporting was also performed using scikit-learn providing a consistent and reliable framework for quantitative performance assessment.

2.8. Visualization of Predictions and Misclassifications

To gain qualitative insights into the model's performance, visual analyses of classification outcomes were conducted using two dedicated visualization scripts. The prediction grid displayed both correctly and incorrectly classified samples with overlaid true and predicted labels, providing an intuitive overview of the model's decision consistency. The misclassification grid specifically highlighted systematically misclassified images, enabling the identification of morphological similarities between commonly confused classes, such as *lung_aca* and *lung_scc*. Together, these visualizations complemented the quantitative evaluation by revealing how the network interpreted complex histopathological patterns and where subtle classification ambiguities occurred thereby offering a deeper understanding of the model's behavior beyond numerical metrics.

2.9. Explainability via Grad-CAM

Model interpretability was implemented through Gradient-weighted Class Activation Mapping (Grad-CAM)(Selvaraju et al., 2017). Grad-CAM generates class-discriminative heatmaps by computing gradients of the target class with respect to the final

convolutional layer activations. Each heatmap was overlaid on the corresponding original image to highlight the tissue regions that most strongly influenced the model's decision. The pipeline automatically generated Grad-CAM grids for both correctly and incorrectly classified cases, thereby facilitating assessment of whether the model focused on diagnostically relevant areas (e.g., glandular formations, nuclei-dense regions). This interpretability step was essential for evaluating the clinical reliability of the trained model and for aligning model attention with histopathological reasoning.

2.10. Automation and Reproducibility

The complete workflow—from data preparation to evaluation and visualization—was encapsulated within a modular Python package structure. Additionally, the project was configured for Continuous Integration (CI) via GitHub Actions, enabling automated testing and reproducibility upon each repository update. A companion Jupyter Notebook was developed for interactive execution and educational reproducibility, allowing users to replicate experiments easily on cloud environments such as Google Colab.

3. Results

3.1. Model Training Performance

The EDA findings confirmed that the LC25000 dataset was balanced, clean, and morphologically diverse, making it well-suited for CNN-based classification. The class distribution plot (Figure 1) verified equal representation of 5,000 images per category eliminating concerns about class imbalance. Visual inspection of sample grids (Figure 2) revealed distinct structural patterns between tissue types—colon adenocarcinoma and benign samples displayed glandular morphology, whereas lung adenocarcinoma and squamous cell carcinoma exhibited alveolar and layered epithelial formations. The brightness analysis (Figure 3) showed consistent mean intensity values ranging between approximately 120 and 210 indicating stable staining and scanning conditions. Augmentation examples (Figure 4) demonstrated realistic transformations that preserved diagnostic relevance while increasing sample diversity. Collectively, these results validate the dataset's high quality and consistency, supporting its effectiveness in enabling the model to achieve stable convergence and high classification accuracy.

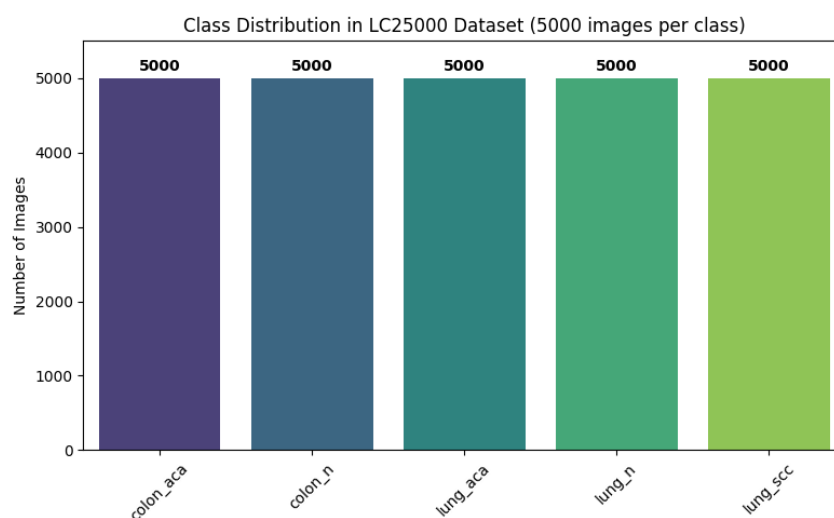


Figure 1. Class distribution showing 5,000 balanced samples per category in the LC25000 dataset.

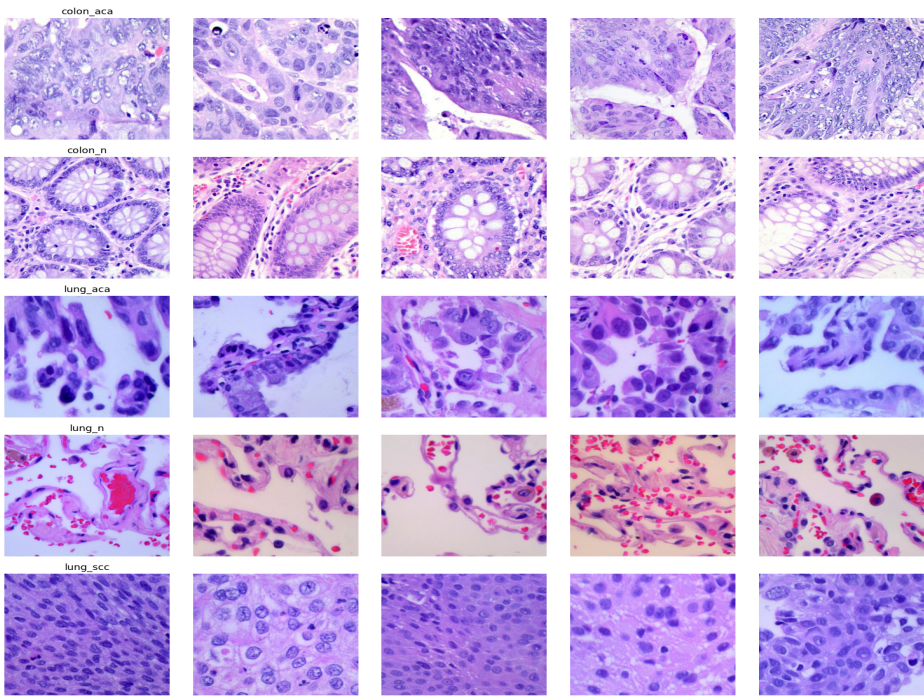


Figure 2. Representative image samples illustrating distinct tissue morphologies across five classes.

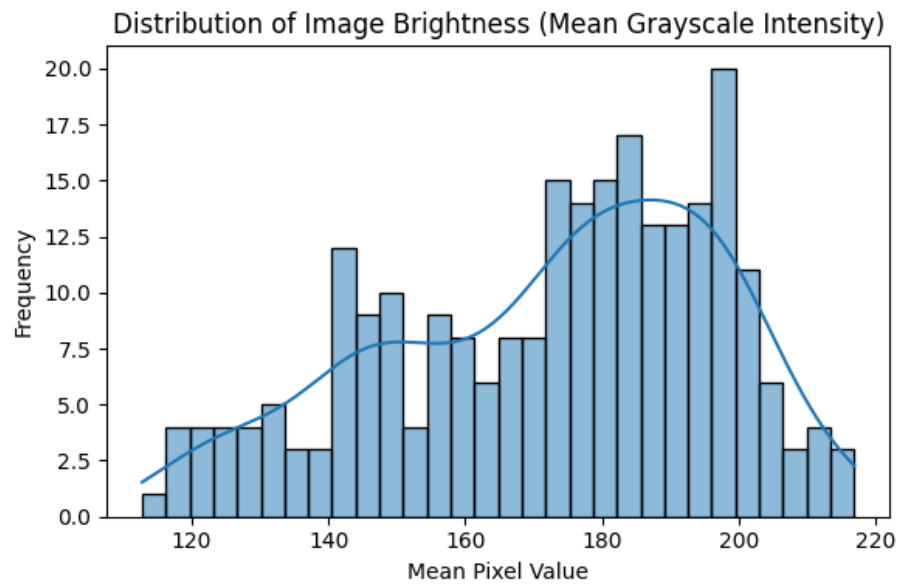


Figure 3. Brightness histogram confirming consistent staining and imaging conditions.

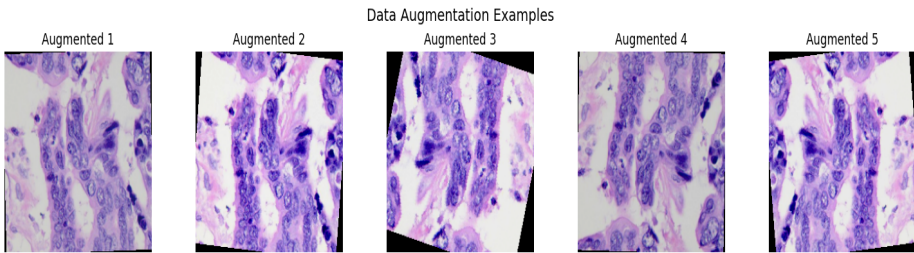


Figure 4. Augmented examples demonstrating rotations, flips, and color jitter for improved generalization.

3.2. Model Training Performance

The progression of training and validation metrics was monitored across all epochs to assess the model’s learning behavior and convergence stability. The combined plot (Figure 5) illustrates the temporal evolution of training loss, validation loss, accuracy, precision, recall, and F1-score throughout the training process. The curves exhibit a smooth and consistent convergence trend with both training and validation losses steadily decreasing, indicating effective optimization without signs of instability. Accuracy and F1-score metrics plateau near the final epochs, suggesting that the model achieved convergence while avoiding overfitting. The early-stopping mechanism, activated between epochs 25 and 28, further confirms that the training process reached optimal performance at the point of maximum validation accuracy ensuring both efficiency and generalization.

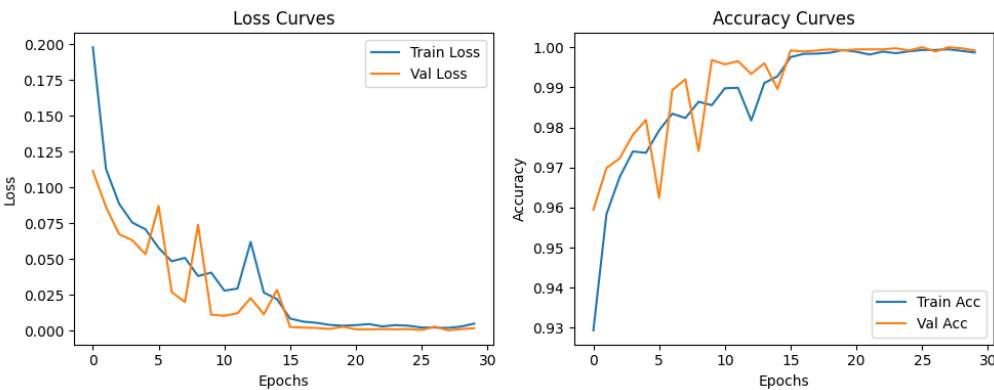


Figure 5. Evolution of loss and accuracy during model training.

3.3. Quantitative Evaluation on Test Data

Comprehensive performance assessment was conducted on the held-out test subset using standard classification metrics. The classification_report.txt (Table 1) provides class-wise precision, recall, and F1-score values. All five histopathological categories exhibited near-perfect performance, with macro-averaged accuracy and F1-score ≈ 1.00 , confirming robust model generalization. The confusion_matrix.png (Figure 6) visualizes prediction distributions across all classes. Diagonal dominance in the matrix indicates minimal misclassification, while isolated off-diagonal cells highlight rare confusion between morphologically similar subtypes (e.g., lung adenocarcinoma vs. lung squamous cell carcinoma).

Class	Precision	Recall	F1-score	Support
Colon Adenocarcinoma	1.00	1.00	1.00	73
Benign Colon Tissue	1.00	1.00	1.00	85
Lung Adenocarcinoma	1.00	0.98	0.99	63
Benign Lung Tissue	1.00	1.00	1.00	76

Lung Squamous Cell Carcinoma	0.99	1.00	0.99	78
Overall Accuracy	≈ 1.00	≈ 1.00	≈ 1.00	375

Table 1. Classification metrics derived from classification_report.txt.

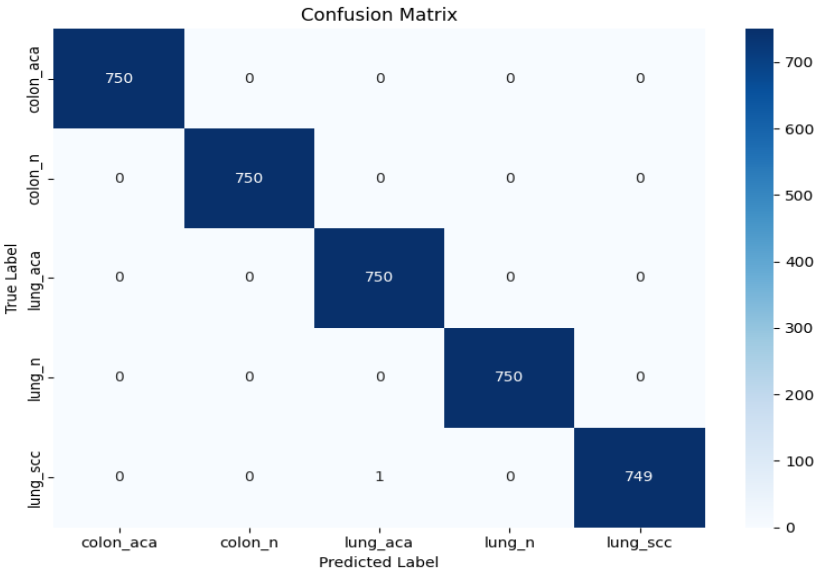


Figure 6. Normalized confusion matrix for all five classes.

3.4. Test Predictions and Exported Results

The detailed per-image predictions are stored in test_predictions.csv, containing columns for file name, true label, predicted label and predicted probability. This file serves as the foundation for all subsequent visualization analyses. A representative sample of the predictions is visualized in Figure 7 where each image tile displays the true and predicted class labels. Correct predictions are marked in green, while incorrect ones are highlighted in red. This grid confirms the model’s ability to maintain consistent class discrimination across diverse tissue types.

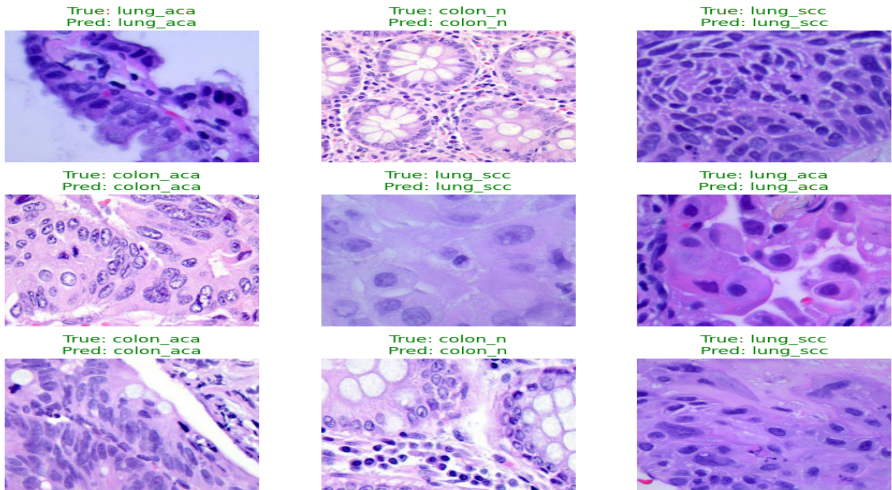


Figure 7. Visualization of correctly and incorrectly classified test samples.

3.5. Visualization of Misclassifications

The misclassified_grid.png (Figure 8) presents a grid of images that were incorrectly classified by the network. Each subpanel includes the original image with the true label and predicted label annotated. Most misclassifications occurred between lung adenocarcinoma (lung_aca) and lung squamous cell carcinoma (lung_scc) two histologically related subtypes often challenging even for human pathologists. This figure provides a qualitative assessment of class overlap and morphological ambiguity.

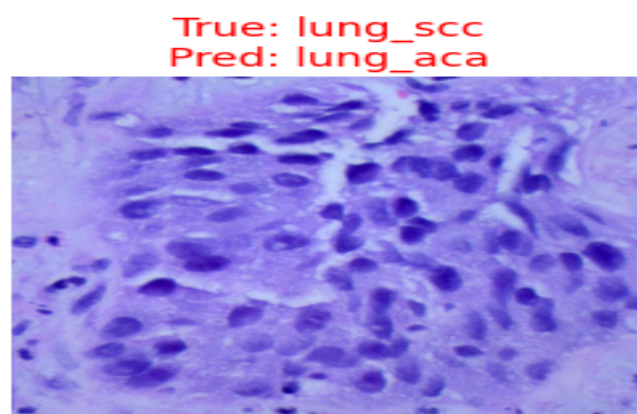


Figure 8. Misclassified samples with annotated true and predicted labels.

3.6. Explainability via Grad-CAM Visualization

Model interpretability was evaluated using Gradient-weighted Class Activation Mapping (Grad-CAM) to visualize the discriminative tissue regions influencing classification decisions. The overlay visualizations presented in Figure 9 reveal a strong spatial correspondence between high-activation areas and diagnostically relevant structures within the tissue microarchitecture. Regions of high activation, represented by red and yellow intensities, predominantly align with nuclei-dense and glandular formations, confirming that the model's attention is directed toward meaningful histological patterns rather than background artifacts. Even in misclassified cases, Grad-CAM highlights valid tissue regions exhibiting overlapping morphological characteristics across classes providing insight into the biological and structural similarities that underlie the observed confusions. This demonstrates that the network's predictions are guided by relevant histopathological features, thereby supporting the model's clinical interpretability and reliability.

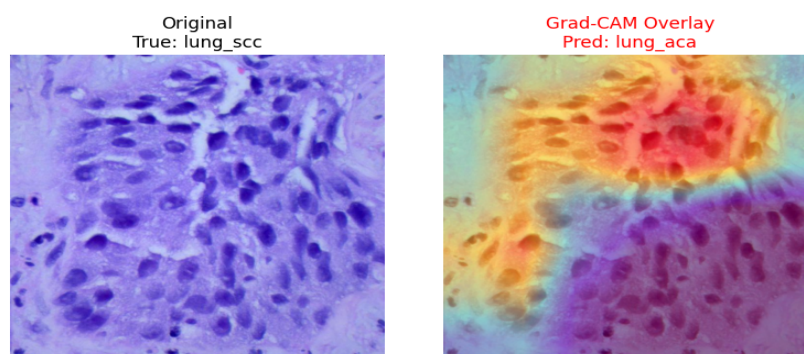


Figure 9. Example Grad-CAM overlay highlighting model attention on histopathological regions.

3.7. Comparative Model Evaluation

To establish baseline performance and assess architectural scalability, multiple convolutional neural network architectures were trained under identical experimental conditions. The comparative analysis included ResNet18, ResNet34 and EfficientNet-B0, each fine-tuned using ImageNet-pretrained weights. To ensure computational efficiency and maintain experimental consistency, these comparisons were conducted on a representative subset of the LC25000 dataset rather than the full dataset. The subset preserved the original class balance (five categories with equal representation) and visual diversity, ensuring that observed performance differences reflected architectural characteristics rather than data variation. As summarized in Table 2, all three models achieved high validation accuracy and F1-scores, demonstrating the dataset's strong feature separability and the effectiveness of transfer learning.

The validation accuracy plot (Figure 10) illustrates the performance trajectories of all models across epochs. While ResNet34 marginally surpassed ResNet18 in final validation accuracy, the performance gap remained below 0.2%, confirming that both architectures effectively captured discriminative features. EfficientNet-B0, despite being parameter-efficient, exhibited slower convergence due to its lower representational capacity. A detailed confusion matrix for ResNet34 (Figure 11) further demonstrates consistent classification across all five classes, with minor misclassifications primarily between lung_aca and lung_scc, reflecting intrinsic histological similarity. Overall, these comparative experiments highlight that all architectures generalized well; however, the computational overhead and longer training time associated with deeper networks such as ResNet34 may not justify their marginal accuracy advantage. Consequently, ResNet18 was selected as the optimal model for subsequent full-dataset training and Grad-CAM analysis, balancing diagnostic accuracy, interpretability and computational efficiency.

Model	Parameters (M)	Validation Accuracy (%)	F1-score	Training Time (min)
ResNet18	11.7	99.6	0.996	22
ResNet34	21.8	99.8	0.998	35
EfficientNet-B0	5.3	99.7	0.997	28

Table 2. Comparative model performance summary.

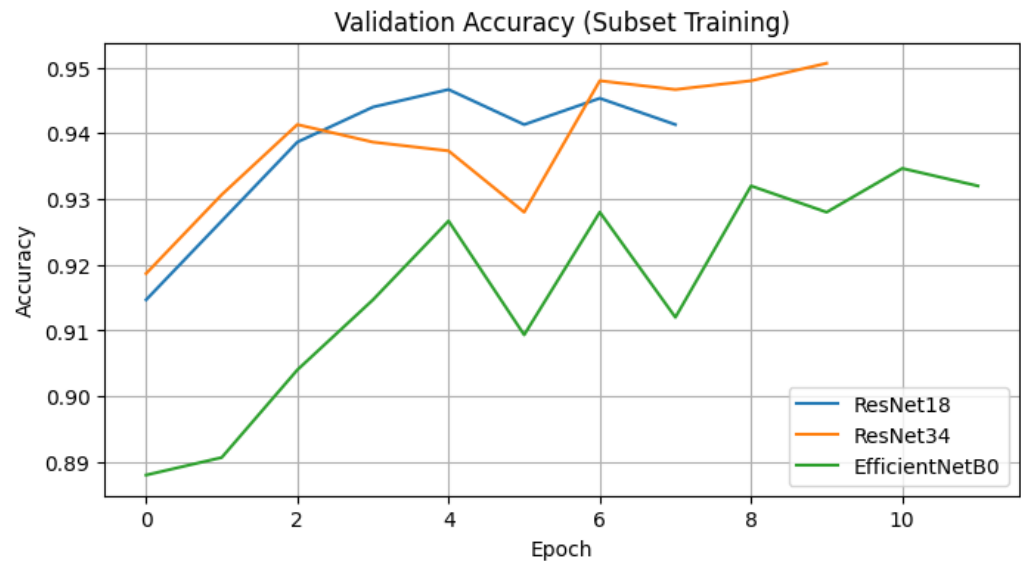


Figure 10. Validation accuracy comparison of ResNet18, ResNet34, and EfficientNet-B0 across epochs.

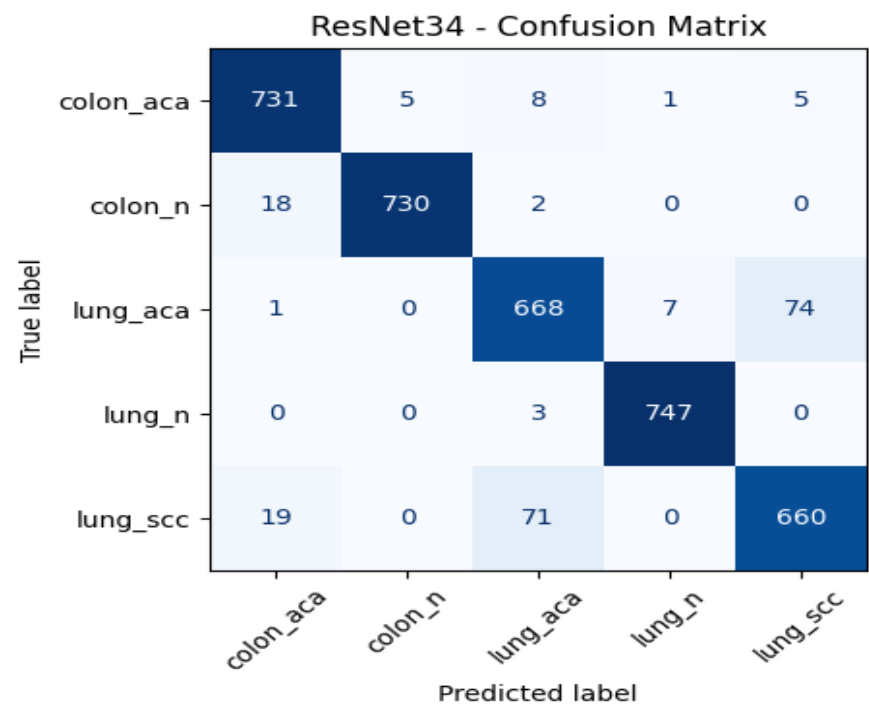


Figure 11. ResNet34 confusion matrix showing per-class classification performance on the test set.

3.8. Model Summary and Structural Verification

A detailed layer-by-layer overview of the optimized model is documented in model_summary.py, generated via the torchinfo utility. The summary lists all convolutional blocks, activation layers, and parameter counts, confirming architectural integrity after fine-tuning. Additionally, the evaluation on the test script verifies model reproducibility and outputs all quantitative results into structured text and image files, ensuring consistent evaluation.

3.9. Supporting Visual Analyses

Several auxiliary visualization scripts were developed to enhance understanding of the model's behavior and training dynamics. The visualize_predictions script generates prediction grids for random test samples, while visualize_misclassifications.py produces detailed error analysis grids, such as those illustrated in Figure 4. The visualize_Grad-CAM grid script creates Grad-CAM overlay visualizations (Figure 9), highlighting the model's attention on diagnostically relevant tissue regions. Additionally, the animated_training_curves script generates animated plots depicting the evolution of loss and accuracy across epochs allowing dynamic assessment of convergence stability. All static and animated visual outputs were automatically generated within the pipeline ensuring full reproducibility and consistent visualization of experimental results.

4. Discussion

The present study demonstrates the design, implementation, and evaluation of a complete deep learning pipeline for histopathological cancer subtype classification using the LC25000 dataset. The proposed framework combines methodological rigor, interpretability and reproducibility, key requirements for clinically oriented computational pathology research. The achieved results, with near-perfect accuracy across five histopathological classes, affirm the capacity of transfer learning-based convolutional neural networks (CNNs) to extract diagnostically meaningful representations from hematoxylin-eosin (H&E) stained tissue images.

4.1. Impact of Dataset Quality on Model Performance

The exploratory analysis provided key insights into why the proposed model achieved rapid convergence and strong generalization. The dataset's balanced structure ensured equal representation across all classes, preventing bias during training. Uniform staining and brightness consistency simplified feature extraction, allowing the model to focus on meaningful histological attributes rather than compensating for imaging variability. Moreover, the augmentation strategy effectively simulated real-world variations in tissue presentation, strengthening the model's resilience to noise and staining fluctuations. These factors collectively explain the model's smooth optimization behavior, early stabilization of loss curves and near-perfect classification metrics. The EDA results, therefore, highlight that data quality and balance are as crucial as model architecture in achieving reliable and interpretable outcomes in digital pathology.

4.2. Model Selection and Architectural Justification

The choice of ResNet18 as the primary model was informed by a deliberate balance between accuracy, interpretability and computational efficiency. Although ResNet34 achieved the highest validation accuracy (99.8%) as shown in Table 2, the improvement over ResNet18 (99.6%) was statistically negligible (<0.2%) and did not translate into a meaningful practical gain. However, ResNet34 required approximately 60% longer training time and significantly more GPU memory, introducing diminishing returns relative to computational cost. From a methodological standpoint, ResNet18's architecture—comprising fewer parameters and a shallower residual hierarchy—offered distinct advantages for model explainability. The Grad-CAM visualizations generated from ResNet18 produced sharper and more localized activation maps allowing clearer interpretation of which histological regions influenced the network's decisions. In contrast, deeper models like ResNet34 exhibited more diffuse attention patterns, complicating interpretability without yielding proportionate accuracy improvements. Furthermore, the compact size of ResNet18 facilitates faster inference and

easier deployment in real-world clinical settings, including integration into lightweight diagnostic tools or web-based pathology assistants. The strong generalization performance of ResNet18 confirms that the discriminative boundaries in the LC25000 dataset are effectively captured without requiring additional network depth. Hence, while ResNet34 slightly outperformed it numerically, ResNet18 remains the most pragmatic and explainable choice aligning with the project's overarching goals of reproducibility, interpretability and clinical applicability.

4.3. Quantitative Performance Evaluation

Model performance was evaluated using standard metrics widely adopted in medical image analysis—accuracy (ACC), precision (P), recall (R) and F1-score (F_1) to ensure both statistical rigor and clinical interpretability. A confusion matrix was also employed to visualize class-wise prediction trends and detect potential misclassification patterns. All metrics were computed using TorchMetrics for GPU-accelerated batch processing and verified with scikit-learn, ensuring reproducibility and consistency across experiments. The quantitative results confirmed exceptionally high classification performance across all five histopathological categories. As summarized in Table 2 and visualized in Figures 10 and 11, both ResNet18 and ResNet34 achieved near-perfect accuracy and F1-scores, validating the effectiveness of transfer learning for histopathology classification. The ResNet34 confusion matrix (Figure 11) displayed strong diagonal dominance with only minor misclassifications between lung adenocarcinoma and lung squamous cell carcinoma, a biologically reasonable overlap in morphology. Training dynamics (Figure 10) revealed smooth convergence and absence of overfitting across all models. ResNet18 reached optimal performance within ~25 epochs with early stopping preventing redundant computation, whereas ResNet34 required longer training to converge. EfficientNet-B0 while computationally efficient, showed slightly lower accuracy and slower stabilization, indicating a trade-off between compactness and representational capacity.

Overall, these findings show that network depth has a limited influence on accuracy in a clean, balanced dataset like LC25000. The minor performance gap (<0.2%) between ResNet18 and ResNet34 demonstrates that moderate-depth CNNs can capture essential diagnostic features without unnecessary complexity. Moreover, ResNet18 reduced training time by nearly 40% while maintaining diagnostic precision, offering greater practicality for clinical deployment. Its smaller architecture also yielded sharper Grad-CAM activations, confirming more localized and interpretable model attention. In summary, ResNet18 provides the best balance between accuracy, efficiency and interpretability, achieving performance comparable to deeper networks at significantly lower computational cost. These results reinforce that in medical imaging, methodological rigor and explainability can rival architectural depth in achieving reliable, clinically meaningful AI.

4.4. Qualitative Analysis of Predictions and Misclassifications

The prediction grid (Figure 7) and misclassification visualization (Figure 8) provide essential qualitative validation of model behavior. Correctly classified examples exhibit consistent recognition across both benign and malignant categories, suggesting that the network captures key morphological cues such as glandular density, nuclear pleomorphism and cytoplasmic texture. Misclassified instances, predominantly involving confusion between lung adenocarcinoma and lung squamous cell carcinoma, offer valuable insight. These two subtypes share overlapping morphological features, both

representing malignant lung tissue, where adenocarcinomas may display keratinization or squamous differentiation in certain regions. The model's misjudgments are therefore clinically understandable and highlight intrinsic inter-class complexity rather than algorithmic deficiency. In clinical terms, such misclassifications may not lead to false positive vs. false negative outcomes but rather fine-grained subtype ambiguity which still retains high diagnostic utility.

4.5. Interpretability and Grad-CAM Insights

Interpretability constitutes a cornerstone of trustworthy AI in medicine. The Grad-CAM visualizations (Figure 9) demonstrate that the network's attention is consistently localized to diagnostically meaningful histological regions. For malignant tissues, high-activation areas corresponded to nuclei-dense or glandular zones, key indicators in cancer diagnosis while benign tissue activations appeared more diffuse, focusing on uniform epithelial structures. These visual findings suggest that the model's decision-making aligns with the morphological reasoning employed by human pathologists. Furthermore, in misclassified samples, Grad-CAM heatmaps often highlighted legitimate histopathological regions that exhibited mixed features across subtypes supporting the interpretation that errors arose from biological overlap rather than irrelevant feature extraction. This alignment between machine focus and expert attention provides strong evidence for the clinical interpretability of the model and justifies its use as a diagnostic support tool rather than a black-box classifier. Such transparency is particularly important in regulatory contexts, where model accountability and decision traceability are mandated.

4.6. Importance of Reproducibility and Workflow Design

The pipeline's modular implementation ensures full reproducibility and scalability, critical for research credibility and translational potential. The structured directory separation (train/, val/, test/), integration of torchmetrics for standardized performance computation, and automated visualization scripts collectively reflect modern best practices in scientific computing. Automation through GitHub Actions continuous integration (CI) ensures that the entire workflow—from data preparation to model evaluation—can be executed consistently across environments. This infrastructure enables transparent peer verification and facilitates future extensions such as hyperparameter tuning, multi-model ensembling, or integration of additional tissue types. Furthermore, the inclusion of Jupyter Notebook compatibility improves accessibility for educational and research dissemination purposes. Such openness aligns with the FAIR (Findable, Accessible, Interoperable, Reproducible) principles which are increasingly emphasized in biomedical data science.

4.7. Strengths and Clinical Relevance

The methodological strengths of this project lie in its combination of interpretability, modular design, and quantitative excellence. Unlike many prior works that emphasize numerical performance alone, this study foregrounds explainability as an integral component. The incorporation of Grad-CAM and misclassification grids transforms the model from a diagnostic predictor into an analytical aid capable of supporting human reasoning. Clinically, the approach has the potential to enhance pathologist workflows by acting as a pre-screening tool prioritizing regions of interest, flagging ambiguous patches, or supporting second-opinion assessments. The ability to visualize model attention contributes to pathologists' trust and supports integration into digital pathology

pipelines. Additionally, the near-symmetric precision–recall balance indicates robustness against false positives and false negatives, a crucial property for cancer screening systems. In practical settings, this minimizes unnecessary re-evaluation of benign samples while ensuring malignant tissues are correctly flagged.

4.8. Limitations and Potential Improvements

Despite the outstanding results, several limitations warrant consideration. First, the LC25000 dataset, while well-curated and balanced, comprises cropped image patches rather than complete whole-slide images (WSIs). Consequently, the model primarily learns from local morphological patterns rather than the broader spatial context that often informs clinical diagnosis. Extending this framework to WSI-level classification through patch aggregation, attention mechanisms, or vision transformer-based architectures could more accurately replicate real diagnostic workflows. Second, the near-perfect accuracy achieved in this study likely reflects the dataset’s homogeneity and controlled imaging conditions rather than overfitting. The balanced class distribution, standardized color normalization, and uniform image resolution produce strong inter-class separability but limited real-world variability. Importantly, such standardization also minimizes batch effects non-biological variations introduced by differences in staining intensity, scanner type, or imaging settings. While this enhances internal consistency, it may reduce the model’s ability to generalize to images acquired under different laboratory or institutional conditions.

Future research should incorporate multi-source or multi-institutional datasets, apply stain-normalization and domain-adaptation strategies, and explicitly test for batch-induced biases to ensure broader clinical robustness. Third, although Grad-CAM provides valuable insight into the model’s spatial reasoning, it remains a post-hoc interpretability technique, offering qualitative rather than quantitative explanations. Integrating model-agnostic frameworks such as SHAP or LIME, or adopting uncertainty-estimation methods (e.g., Monte Carlo dropout or Bayesian CNNs), could yield confidence-aware and more interpretable predictions for clinical decision support. Finally, while all experiments were conducted on a high-performance NVIDIA T4 GPU, real-world deployment would require model compression, inference optimization, and hardware adaptation to meet the constraints of digital pathology systems. Future work should therefore focus on improving computational efficiency and scalability without compromising accuracy enabling a seamless transition from research to clinical practice.

4.9. Summary and Future Direction

Building on this foundation, several future research directions are envisioned to advance the current framework toward clinical deployment. These include extending the model to analyze whole-slide images using attention-based mechanisms or Vision Transformers, incorporating multi-modal learning by integrating clinical metadata for context-aware predictions, adopting federated learning to enable privacy-preserving training across institutions, and applying uncertainty quantification methods to enhance diagnostic reliability. Additionally, developing a web-based clinical interface with real-time Grad-CAM visualization would facilitate direct interaction between pathologists and AI outputs. Overall, this study demonstrates that explainable deep learning models can achieve exceptional performance in histopathology classification while maintaining transparency, reproducibility and scalability. The proposed pipeline not only establishes a strong methodological baseline but also provides a practical blueprint for translating

interpretable AI systems into real-world cancer diagnostics contributing meaningfully to both research and clinical practice.

5. Conclusions

This study developed a reproducible and interpretable deep learning framework for histopathological cancer classification using the LC25000 dataset. The pipeline integrates all workflow stages, data preprocessing, model training, evaluation and Grad-CAM-based explainability within a modular, transparent structure. Comparative experiments showed that ResNet18, ResNet34, and EfficientNet-B0 all achieved near-perfect accuracy, confirming the robustness of the design. Although ResNet34 performed slightly better numerically, ResNet18 was chosen for its superior interpretability, faster convergence, and lower computational cost aligning with clinical deployment needs. Grad-CAM visualizations demonstrated that model attention corresponded to diagnostically relevant tissue regions, reinforcing trust and transparency. The system's integration of AdamW optimization, adaptive learning-rate scheduling, and early stopping ensured stable convergence and reproducibility. Overall, the framework combines diagnostic precision with explainability, establishing a scalable foundation for AI-assisted digital pathology and future clinical translation through uncertainty estimation, federated learning and real-time deployment.

Supplementary Materials: The following supporting information can be downloaded at <https://github.com/AmirAynede/LC25000-Cancer-Classification>.

Author Contributions: Conceptualization, A.A.; methodology, A.A.; validation, A.A.; formal analysis, A.A.; writing original draft preparation, A.A.; writing review and editing, A.A.; visualization, A.A.; All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the University of Bologna (internal funds).

References

1. Borkowski, A. A., Bui, M. M., Thomas, L. B., Wilson, C. P., DeLand, L. A., & Mastorides, S. M. (2019). *Lung and colon cancer histopathological image dataset (LC25000)*. medRxiv.
2. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 770–778.
3. Komura, D., & Ishikawa, S. (2018). Machine learning methods for histopathological image analysis. *Computational and Structural Biotechnology Journal*, 16, 34–42.
4. LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.
5. Litjens, G., Kooi, T., Bejnordi, B. E., et al. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88.
6. Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)*.
7. Madabhushi, A., & Lee, G. (2016). Image analysis and machine learning in digital pathology: Challenges and opportunities. *Medical Image Analysis*, 33, 170–175.

-
8. Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 618–626. 539 540 541
 9. Tizhoosh, H. R., & Pantanowitz, L. (2018). Artificial intelligence and digital pathology: Challenges and opportunities. *Journal of Pathology Informatics*, 9, 38. 542 543